

OS Architecture, System Calls & Processes - Detailed Lecture Notes

DETAILED COMPREHENSIVE NOTES

Subject: Operating Systems | Chapter: Chapter 1: OS Architecture, System Calls & Process Management | Academic Year: 2026-2027

1. Operating System Architecture & Dual-Mode Operation

An Operating System is an intermediary between computer hardware and user applications, providing hardware abstraction and resource multiplexing.

Dual-Mode Operation protects hardware: User Mode (restricted instructions, cannot directly access I/O or kernel memory) and Kernel Mode (privileged instructions executed by CPU). Transitions occur via software interrupts (System Calls) or hardware interrupts.

2. Process Concept & Process Control Block (PCB)

A process is an active instance of a program in execution. Its address space consists of: Text Segment (compiled code), Data Segment (global/static variables), Heap (dynamically allocated memory), and Stack (local variables, return addresses).

The Process Control Block (PCB) maintains execution context: Process ID (PID), Process State, Program Counter, CPU registers, CPU scheduling info, Memory management info, and I/O status.

- * Process States: New, Ready, Running, Waiting (Blocked), and Terminated.
- * Context Switch: Storing the state of the active process in its PCB and loading the saved state of the newly scheduled process.

3. Process Creation, Termination & Inter-Process Communication (IPC)

In Unix-like systems, processes are created using the `fork()` system call, creating a child duplicate of the parent process. The child receives return value 0, while the parent receives child PID.

Inter-Process Communication mechanisms allow processes to exchange data: Shared Memory (fastest, requires synchronization) and Message Passing (queues, sockets, pipes).